BMC Public Health

RESEARCH

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Long-term exposure to fine particulate matter and its constituents is related to self-reported visual impairment among middle-aged and older adults: a longitudinal study in China

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Abstract

Background Visual impairment (VI) poses an increasing global health burden, yet the long-term effects of fine particulate matter ($PM_{2.5}$) constituents on VI remain unclear. This study aimed to evaluate associations between long-term exposure to $PM_{2.5}$ and its constituents and the risk of VI.

Methods We used data from the nationally representative China Health and Retirement Longitudinal Study (CHARLS, 2011–2018) including adults aged ≥ 45 years. Long-term exposure to $PM_{2.5}$ and its constituents (SO_4^{2-}, NO_3^-, NH_4^+, organic matter (OM), and black carbon (BC)) was obtained from the Tracking Air Pollution in China dataset and linked to participants' residential cities. The 3-year moving average concentration was used as the primary exposure metric to capture cumulative effects, while the 1-year average concentration was applied for sensitivity analysis. VI was defined as self-reported fair or poor vision. Associations were estimated using time-dependent Cox models, with restricted cubic spline (RCS), stratified and interaction analyses to assess nonlinear and subgroup effects.

Results Over 27,443 person-years of follow-up, 3,736 participants developed VI. Per interquartile range (IQR) increase in 3-year moving average concentrations, higher risks of VI were observed for $PM_{2.5}$ (HR = 1.09, 95% CI: 1.03–1.15), SO_4^{2-} (HR = 1.13, 95% CI: 1.05–1.22), NO_3^- (HR = 1.06, 95% CI: 1.00–1.13), NH_4^+ (HR = 1.09, 95% CI: 1.02–1.17), OM (HR = 95% CI: 1.15, 1.07–1.24), and BC (HR = 1.24, 95% CI: 1.14–1.34). RCS analysis revealed non-linear associations for $PM_{2.5}$, NO_3^-, NH_4^+, and OM, whereas SO_4^{2-} and BC showed approximately linear trends. Associations were more pronounced among older adults (> 60 years old), smokers, and those with hypertension.

Conclusion Long-term exposure to $PM_{2.5}$ and its constituents significantly increases the risk of VI among middle-aged and older adults in China, highlighting the need for targeted environmental health policies.

Keywords Black carbon, Fine particulate matter, Time-dependent Cox models, Visual impairment

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Introduction

Visual impairment (VI) has emerged as a pressing global public health issue, imposing considerable social and health burdens worldwide. According to data from the Global Burden of Disease (GBD) studies, the number of individuals affected by moderate and severe VI increased by 133.67% and 147.14%, respectively, between 1990 and 2019. Projection indicated that by 2050, approximately 474 million people worldwide will experience moderate-to-severe VI [1, 2]. The burden of VI extends beyond visual loss itself, encompassing a wide range of physical, psychological, and socioeconomic consequences that substantially affect quality of life and overall well-being. Individuals with VI typically engage in less physical activity—a key determinant of overall health and longevity—and are at higher risk of falls due to impaired mobility and daily functioning [3, 4]. Moreover, VI is closely linked to social isolation, cognitive decline, depression, and anxiety [5–8]. Importantly, VI may not only result from ocular diseases but also from environmental exposures, notably air pollution.

Accumulating evidence suggests that air pollution exposure contributes to the development of various ocular diseases. Previous studies have demonstrated that exposure to ambient particulate matter (PM) can induce ocular surface disorders, including conjunctivitis, keratitis, and dry eye disease [9–11]. Of particular concern is fine particulate matter ($PM_{2.5}$), characterized by its small aerodynamic diameter and large surface area. Recent studies have associated $PM_{2.5}$ exposure with cataract, glaucoma, age-related macular degeneration (AMD), and retinal or choroidal injury [12–15]. Mechanistically, $PM_{2.5}$ may impair ocular tissues through neurotoxicity, cellular pyroptosis, and disruption of retinal pigment epithelial tight junctions, thereby predisposing individuals to glaucoma, optic atrophy, and other retinal diseases [12, 16, 17].

Despite these findings, evidence linking $PM_{2.5}$ exposure to VI remains limited. Moreover, the chemical constituents of $PM_{2.5}$—such as sulfates (SO_4^{2-}), nitrates (NO_3^-), ammonium (NH_4^+), organic matter (OM), and black carbon (BC)—vary substantially in biological toxicity and may differentially affect ocular health. While prior research has explored associations between $PM_{2.5}$ and VI in children, dose–response patterns, and vulnerable subpopulations among adults remain poorly characterized. This study therefore aims to comprehensively investigate the associations between long-term exposure to $PM_{2.5}$ and its constituents and the risk of VI.

Methods

Study design and population

This prospective cohort study utilized data from the China Health and Retirement Longitudinal Study (CHARLS), a nationally representative survey of adults aged 45 years and older. CHARLS employed a multistage probability sampling strategy covering both urban and rural communities across 28 provinces in mainland China [18]. The CHARLS was initiated in 2011 and subsequently followed up in 2013, 2015, 2018, and 2020. The CHARLS was approved by the Biomedical Ethics Review Committee of Peking University (IRB00001052–11015), and written informed consent was obtained from all participants. Due to the lack of VI information in 2020, this study collected extensive sociodemographic, behavioral, and health information at baseline (2011) and during three follow-up waves (2013, 2015, and 2018).

Participants were included if they: (1) were aged ≥ 45 years at baseline; (2) had VI information; and (3) completed at least one follow-up interview. Participants with missing $PM_{2.5}$ exposure estimates, missing key covariates (age), missing baseline VI information, or prevalent VI at baseline were excluded to ensure clear temporal ordering between exposure and outcome. After applying inclusion criteria, 11,831 participants were included, contributing 27,443 person-years of follow-up (Fig. 1).

Exposure assessment

1-year and 3-year moving average concentrations of $PM_{2.5}$ and its constituents (SO_4^{2-}, NO_3^-, NH_4^+, OM, and BC) were obtained from the Tracking Air Pollution in China (TAP) dataset (http://tapdata.org) [19, 20].

TAP integrates multiple data sources, including satellite remote sensing, ground-based monitoring, meteorological reanalysis, and land-use information, and applies ensemble machine learning algorithms (random forest and deep neural networks) to generate 1 km × 1 km gridded annual average estimates of pollutant concentrations across China. We subsequently converted the grid coordinates in the TAP dataset into actual residential addresses and calculated the annual average pollutant concentrations at city level.

Residential information in CHARLS was available only at the city level to protect participant privacy. For each participant, the 3-year moving average concentration, calculated as the mean of the current year and the two preceding years (e.g., 2009–2011 for the 2011–2013 interval), was used as the primary exposure metric. The 3-year moving average concentration smoothed interannual fluctuations, reflects cumulative exposure burden, and better captures the biological latency of chronic ocular effects.

To assess robustness, 1-year average concentrations before the follow-up interval (e.g., 2010–2011 for the 2011–2013 interval) were also analyzed. Overall, pollutant data encompassed 28 provinces and 125 cities across China.

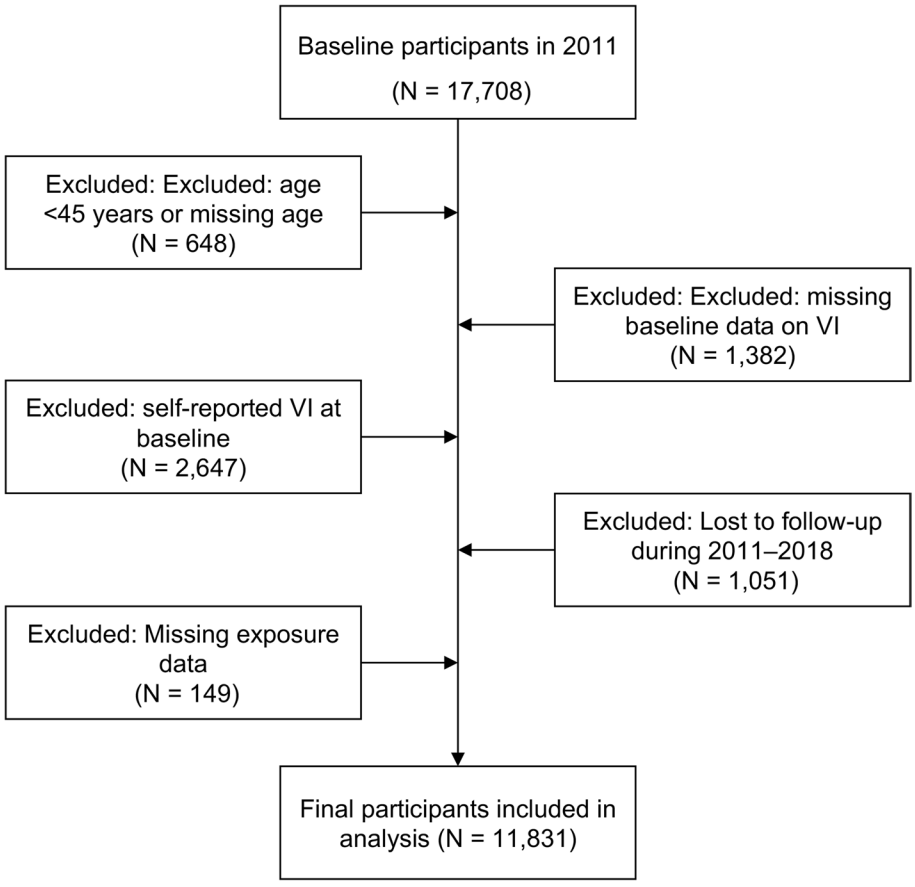


Fig. 1 Flow diagram of participant selection. Abbreviations: VI, visual impairment

Outcome ascertainment

The outcome was VI, determined from self-reported vision assessments in CHARLS. Participants were asked to rate their distance and near vision as "excellent", "very good", "good", "fair", or "poor". Consistent with previous studies, responses of "fair" or "poor" were classified as VI [21–23]. The event time was defined as the wave in which a participant first self-reported VI or lost follow-up.

Covariates

Covariates were selected based on the previous studies [13, 24, 25]. Covariates included age, sex, marital status (married or other), residence (urban or rural), education level (less than primary, primary, middle, or high school and above), personal annual expenditure (CNY) (0–2,915, 2,916–4,807, 4,808–7,343, 7,344–12,108, ≥ 12,109), medical insurance (yes or no), smoking (yes or no), drinking (yes or no), hypertension(yes or no), diabetes(yes or no), dyslipidemia(yes or no), heart disease(yes or no), stroke(yes or no), body mass index (BMI, kg/m^2), systolic and diastolic blood pressure (SBP, DBP, mmHg), total cholesterol (TC, mg/dL), triglycerides (TG, mg/dL), random blood glucose (GLU, mg/dL), and glycated hemoglobin (HbA1c, %). Additionally, province of residence, as a fixed effect to account for regional variations, was incorporated.

Statistical analysis

This study used multiple imputation to handle missing data. Baseline characteristics were compared between participants with and without VI. Continuous variables were summarized as medians with interquartile ranges (IQRs) and compared using the Wilcoxon rank-sum or Kruskal–Wallis tests, while categorical variables were expressed as counts (percentages) and compared using χ^2 or Fisher's exact tests. Associations between long-term $PM_{2.5}$ exposure and the risk of VI were evaluated using time-dependent Cox models. Because $PM_{2.5}$ and its chemical constituents were highly correlated with each other, single-pollutant models were constructed to avoid multicollinearity (Table S1 in Supplementary material). To ensure comparability across pollutants with different concentration ranges, hazard ratios (HRs) were estimated IQR increase in $PM_{2.5}$ and each constituent. Three sequential models were developed: Model 1 adjusted for age, sex, marital status, residence, education level, smoking status, drinking status, and province of residence; Model 2 further adjusted for hypertension, diabetes,

dyslipidemia, heart disease, stroke, personal annual expenditure, and BMI; Model 3 additionally incorporated SBP, DBP, TC, TG, GLU, and HbA1c. To examine nonlinear dose–response relationships, restricted cubic spline (RCS) model with four knots was fitted. Stratified Cox models were applied to assess subgroup-specific associations, and interaction analyses were conducted between air pollutants and potential effect modifiers. All analyses were performed using R software (version 4.4.2; R Foundation for Statistical Computing, Vienna, Austria) with the "survival" package. Statistical significance was defined as a two-sided $P < 0.05$.

Results

Baseline characteristics of participants

At baseline, 11,831 participants aged ≥ 45 years without VI were included, among whom 3,736 (31.6%) reported VI during follow-up (Table S2 in Supplementary Material). The median age of participants was 58 years old, and 54.2% were female. Participants who reported VI were more likely to be reside in urban areas, and have a higher educational level ($P < 0.001$). Participants with VI showed lower prevalence of diabetes and hypertension, but were more often current smokers and drinkers ($P < 0.01$). No significant differences were observed in dyslipidemia and medical insurance coverage. In addition, those who developed VI had slightly higher BMI and HbA1c levels ($P < 0.001$), while lipid factors and blood pressure were similar between two groups. Regarding exposure levels, participants with VI were exposed to slightly lower 1-year average $PM_{2.5}$ but higher 1-year average SO_4^{2-} concentrations at baseline, with consistent patterns for 3-year moving averages ($P < 0.05$).

The relationship between $PM_{2.5}$ and VI

In the time-dependent Cox models, higher 3-year moving average concentrations of $PM_{2.5}$ and its constituents were consistently associated with an increased risk of VI (Table 1). For each interquartile range (IQR) increase in $PM_{2.5}$, the fully adjusted HR for VI was 1.09 (95% CI: 1.03–1.15, $P = 0.003$). The associations were more pronounced for $PM_{2.5}$ constituents, particularly SO_4^{2-} (HR = 1.13, 95% CI: 1.05–1.22), NH_4^+ (HR = 1.09, 95% CI: 1.02–1.17), and OM (HR = 1.15, 95% CI: 1.07–1.24). A modest but significant effect was also observed for NO_3^- (HR = 1.06, 95% CI: 1.00–1.13). Among all constituents, BC exhibited the strongest association with VI (HR = 1.24, 95% CI: 1.14–1.34), highlighting its potential role as a particularly toxic fraction of $PM_{2.5}$.

RCS analysis of $PM_{2.5}$ and its constituents with VI risk

RCS analysis revealed significant dose–response relationships between the 3-year moving average concentrations of $PM_{2.5}$ and its constituents and the risk of VI (Fig. 2). The overall associations were statistically significant for all pollutants (P for overall < 0.05). A nonlinear pattern was observed for $PM_{2.5}$ (P for nonlinearity < 0.001), NO_3^- ($P = 0.004$), NH_4^+ ($P = 0.042$), and OM ($P < 0.001$). In contrast, the relationships for SO_4^{2-} (P for nonlinearity = 0.204) and BC (P for nonlinearity = 0.253) appeared approximately linear, with a gradual rise in risk across the exposure range.

Categorical analysis of 3-year average $PM_{2.5}$ and its constituents and VI risk

To further validate the nonlinear exposure–response patterns indicated by the RCS analysis, pollutant concentrations were categorized into tertiles, and HRs for each tertile were estimated using time-dependent Cox models (Table 2). For most $PM_{2.5}$ constituents, a nonlinear trend was observed: compared with the lowest tertile (T1), the second tertile (T2) generally showed slightly reduced or nonsignificant risks, whereas the third tertile (T3) was associated with elevated VI risk. The cutoff values for tertiles were presented in Table S3 in Supplementary Material.

Specifically, higher risks of VI were found in the top tertile of SO_4^{2-} (HR = 1.26, 95% CI: 1.08–1.48, $P = 0.005$), NH_4^+ (HR = 1.30, 95% CI: 1.11–1.53, $P = 0.001$), and BC (HR = 1.34, 95% CI: 1.16–1.55, $P < 0.001$), while the corresponding T2 levels were below unity but not statistically significant. These results support the nonlinear

Table 1 Associations between 3-year average concentrations of $PM_{2.5}$ and its constituents and risk of VI

Pollutants (per 1 IQR increase)	IQR (μg/m³)	Model 1		Model 2		Model 3	
		HR (95%CI)	*P*	HR (95%CI)	*P*	HR (95%CI)	*P*
$PM_{2.5}$	21.97	1.09 (1.03–1.15)	**0.002**	1.09 (1.03–1.15)	**0.003**	1.09 (1.03–1.15)	**0.003**
SO_4^{2-}	5.02	1.13 (1.05–1.22)	**0.001**	1.13 (1.05–1.22)	**0.001**	1.13 (1.05–1.22)	**0.001**
NO_3^-	6.00	1.07 (1.00–1.13)	**0.037**	1.06 (1.00–1.13)	**0.049**	1.06 (1.00–1.13)	**0.046**
NH_4^+	4.07	1.09 (1.02–1.17)	**0.008**	1.09 (1.02–1.17)	**0.012**	1.09 (1.02–1.17)	**0.011**
OM	6.25	1.16 (1.07–1.25)	**< 0.001**	1.15 (1.06–1.24)	**< 0.001**	1.15 (1.07–1.24)	**< 0.001**
BC	1.18	1.25 (1.15–1.35)	**< 0.001**	1.24 (1.14–1.34)	**< 0.001**	1.24 (1.14–1.34)	**< 0.001**

$P < 0.05$ indicated statistical significance and font was bolded

Abbreviations: *HR* Hazard ratio, *CI* Confidence interval, *$PM_{2.5}$* Particulate matter ≤ 2.5 μm in aerodynamic diameter, *SO_4^{2-}* Sulfates, *NO_3^-* Nitrates, *NH_4^+* Ammonium, *OM* Organic matter, *BC* Black carbon, *VI* Visual impairment

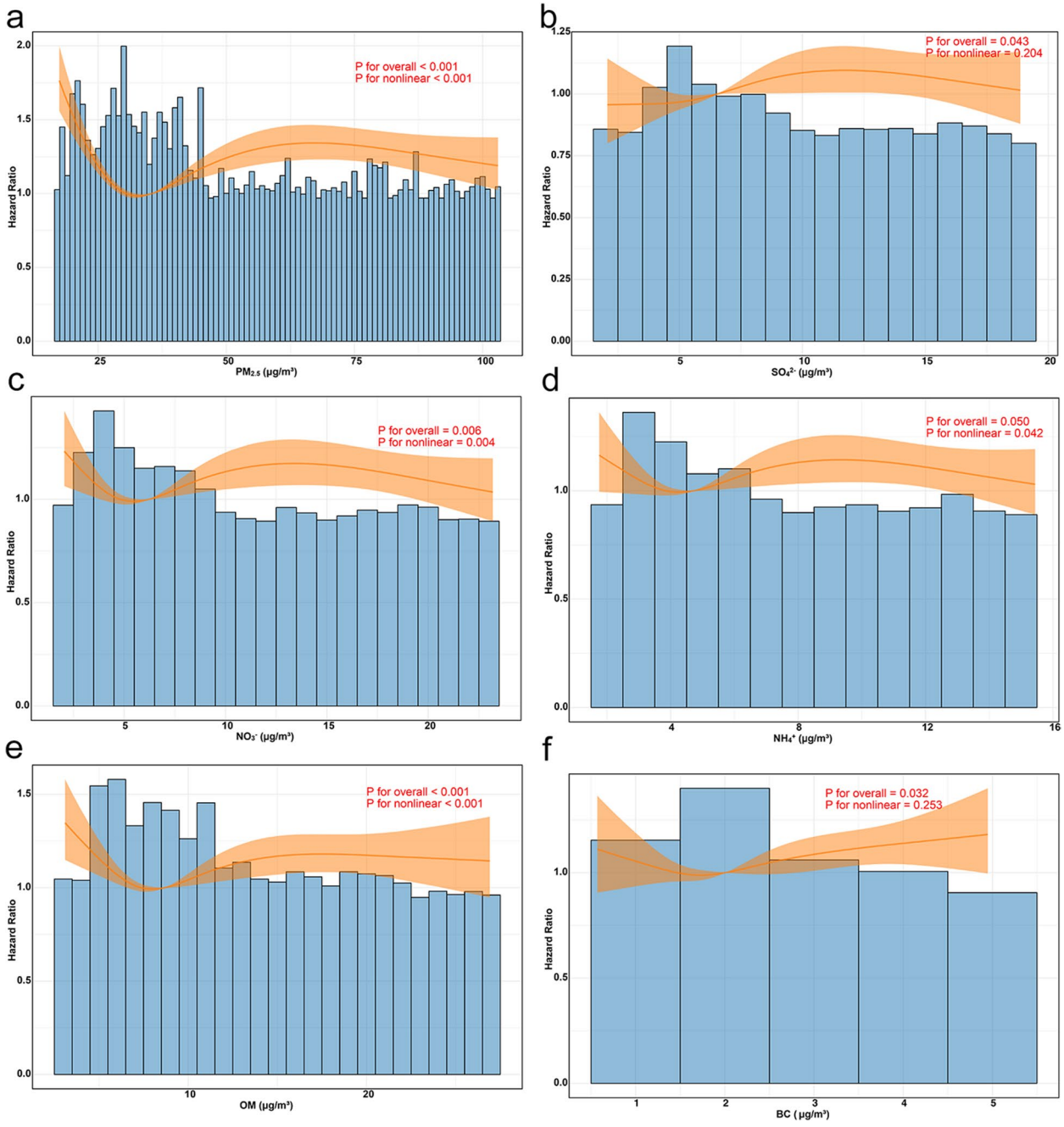


Fig. 2 Dose–response relationships between 3-year average concentrations of $PM_{2.5}$ and its constituents and risk of VI based on RCS models. **a** $PM_{2.5}$; **b** SO_4^{2-};**c** NO_3^-;**d** NH_4^+; **e** OM; **f** BC. RCS models were adjusted for all covariates in Model 3. The solid orange lines represent HR, and shaded areas indicate 95% confidence intervals. Histograms show the distribution of participants across pollutant concentrations. *P* for overall association and *P* for nonlinearity are shown in each panel. Abbreviations: $PM_{2.5}$, particulate matter ≤ 2.5 μm in aerodynamic diameter; SO_4^{2-}, sulfates; NO_3^-, nitrates; NH_4^+, ammonium; OM, organic matter; BC, black carbon

dose–response relationships observed in RCS models, suggesting that the adverse effects of $PM_{2.5}$ constituents become pronounced at higher exposure levels.

Stratified and interaction analyses

Stratified analysis based on 3-year moving average concentrations of $PM_{2.5}$ and its constituents revealed differential susceptibility to VI across population subgroups.

For $PM_{2.5}$, SO_4^{2-}, and NO_3^-, stronger associations were observed in several subgroups, particularly among older adults (> 60 years old), participants with hypertension, current smokers, and non-drinkers. The associations also tended to be more evident in those with primary

Table 2 Associations between tertiles of 3-year average concentrations of $PM_{2.5}$ and its constituents and risk of VI

Pollutant tertiles (reference: T1)	Model 1		Model 2		Model 3	
	HR (95% CI)	*P*	HR (95% CI)	*P*	HR (95% CI)	*P*
$PM_{2.5}$						
T2	0.91 (0.82–1.01)	0.088	0.90 (0.81–1.00)	0.058	0.90 (0.81–1.01)	0.067
T3	1.10 (0.97–1.25)	0.154	1.09 (0.96–1.25)	0.177	1.10 (0.96–1.25)	0.155
SO_4^{2-}						
T2	1.12 (0.98–1.28)	0.092	1.11 (0.98–1.27)	0.110	1.11 (0.98–1.27)	0.110
T3	1.28 (1.09–1.50)	**0.003**	1.26 (1.07–1.48)	**0.005**	1.26 (1.08–1.48)	**0.005**
NO_3^-						
T2	0.92 (0.82–1.04)	0.200	0.92 (0.82–1.04)	0.178	0.93 (0.82–1.05)	0.231
T3	0.99 (0.85–1.15)	0.872	0.99 (0.85–1.15)	0.847	0.99 (0.85–1.15)	0.892
NH_4^+						
T2	1.02 (0.90–1.15)	0.771	1.01 (0.89–1.14)	0.891	1.01 (0.90–1.15)	0.833
T3	1.32 (1.12–1.55)	**0.001**	1.30 (1.10–1.52)	**0.002**	1.30 (1.11–1.53)	**0.001**
OM						
T2	0.99 (0.87–1.12)	0.827	0.98 (0.86–1.11)	0.722	0.98 (0.86–1.12)	0.766
T3	1.15 (1.00–1.33)	0.053	1.14 (0.99–1.32)	0.073	1.14 (0.99–1.32)	0.068
BC						
T2	0.99 (0.87–1.12)	0.880	0.98 (0.87–1.12)	0.805	0.98 (0.87–1.11)	0.788
T3	1.35 (1.17–1.57)	**<0.001**	1.34 (1.16–1.54)	**<0.001**	1.34 (1.16–1.55)	**<0.001**

P < 0.05 indicated statistical significance (bolded)

Abbreviations: *T1* First tertile, *T2* Second tertile, *T3* Third tertile, *Ref* Reference, *HR* Hazard ratio, *CI* Confidence interval, *$PM_{2.5}$* Particulate matter ≤ 2.5 μm in aerodynamic diameter, *SO_4^{2-}* Sulfates, *NO_3^-* Nitrates, *NH_4^+* Ammonium, *OM* Organic matter, *BC* Black carbon

or middle-school education and among individuals with higher personal annual expenditure (Q4). For example, per 1 IQR increase in SO_4^{2-} was associated with a 20% higher risk of VI among participants with hypertension (HR 1.20, 95% CI 1.07–1.34, *P* = 0.002), and a 24% higher risk among those aged > 60 years (HR 1.24, 95% CI 1.10–1.39, *P* < 0.001) (Table S4 in Supplementary Material).

For NH_4^+, OM, and BC, subgroup patterns were consistent but the effect was larger, especially for BC. Elevated risks were prominent among older adults (> 60 years old), hypertensive and dyslipidemic participants, current smokers, and non-drinkers; the associations were also evident across residential settings and educational strata. Notably, BC showed the strongest effects—for instance, per 1 IQR increase in BC was associated with HR 1.42 (95% CI 1.25–1.62, *P* < 0.001) among those aged > 60 years and HR 1.78 (95% CI 1.19–2.67, *P* = 0.005) among participants without medical insurance; significant elevations were also seen in the Q4 expenditure group (HR 1.47, 95% CI 1.23–1.75, *P* < 0.001) (Table S5 in Supplementary Material).

No substantial effect modification was detected in different groups (*P* for interaction > 0.05), suggesting that the observed associations were broadly consistent across all groups (Table 3).

Sensitivity analysis

As a sensitivity analysis, we used the 1-year average concentrations to evaluate the robustness of the associations. The findings were largely consistent with the main analyses, supporting the stability of the results.

Per IQR increase, higher 1-year average concentration of $PM_{2.5}$ (IQR = 19.83 μg/m^3) were significantly associated with a higher risk of VI (HR = 1.09, 95% CI 1.04–1.15). Similar associations were observed for its constituents, including SO_4^{2-} (4.50 μg/m^3; HR = 1.12, 95% CI 1.05–1.20), NO_3^- (5.73 μg/m^3; HR = 1.07, 95% CI 1.01–1.14), NH_4^+ (3.92 μg/m^3; HR = 1.09, 95% CI 1.02–1.17), OM (5.69 μg/m^3; HR = 1.15, 95% CI 1.07–1.24), and BC (1.07 μg/m^3; HR = 1.21, 95% CI 1.12–1.30) (Table 4).

RCS analysis indicated significant nonlinear dose–response relationships for $PM_{2.5}$, SO_4^{2-}, NO_3^-, and OM (*P* for nonlinearity < 0.05). These pollutants showed an initial flattening or mild decline in VI risk at lower concentrations, followed by a pronounced increase in risk at higher exposure levels. In contrast, NH_4^+ and BC exhibited approximately linear dose–response patterns, with risk rising steadily across the exposure range (Figure S1 in Supplementary Material).

Consistent with the RCS results, tertile analysis demonstrated that participants in the highest tertile had significantly greater risks of VI for $PM_{2.5}$ (HR = 1.16, 95% CI 1.02–1.32) and SO_4^{2-} (HR = 1.23, 95% CI 1.05–1.44), whereas associations for NO_3^-, NH_4^+, OM, and BC were weaker and did not reach statistical significance (Table S6 in Supplementary Material).

Stratified analysis showed generally consistent associations across demographic and clinical subgroups (Tables S7–S8 in Supplementary Material). The associations were

Table 3 Interaction effects between 3-year moving average $PM_{2.5}$ and its constituents and individual characteristics on the risk of VI

Covariate	*P* for interaction					
	$PM_{2.5}$ (μg/m³)	SO_4^{2-} (μg/m³)	NO_3^- (μg/m³)	NH_4^+ (μg/m³)	OM (μg/m³)	BC (μg/m³)
Age	0.312	0.173	0.408	0.282	0.204	0.090
Sex	0.593	0.516	0.600	0.495	0.600	0.551
Marital status	0.303	0.517	0.381	0.377	0.340	0.419
Residence	0.621	0.846	0.600	0.655	0.724	0.799
Education level	0.813	0.975	0.921	0.982	0.919	0.958
Diabetes	0.378	0.319	0.538	0.376	0.452	0.196
Hypertension	0.130	0.442	0.409	0.463	0.216	0.234
Dyslipidemia	0.498	0.420	0.396	0.372	0.504	0.532
Smoking, now	0.215	0.131	0.179	0.154	0.426	0.358
Drinking, now	0.811	0.806	0.812	0.986	0.843	0.995
Heart diseases	0.328	0.189	0.312	0.264	0.232	0.210
Stroke	0.508	0.474	0.373	0.428	0.634	0.722
Medical insurance	0.519	0.341	0.416	0.358	0.523	0.358
Personal annual expenditure (CNY)	0.337	0.433	0.221	0.376	0.454	0.751

P values are from interaction terms between each pollutant and covariate in time-dependent Cox models adjusted for all covariate in Model 3

Abbreviations: $PM_{2.5}$ Particulate matter ≤ 2.5 μm in aerodynamic diameter, SO_4^{2-} Sulfates, NO_3^-Nitrates, NH_4^+ Ammonium, *OM* Organic matter, *BC* Black carbon, *VI* Visual impairment

Table 4 Associations between 1-year moving average concentrations of $PM_{2.5}$ and its constituents and VI

Pollutants (per 1 IQR increase)	IQR (μg/m³)	Model 1		Model 2		Model 3	
		HR (95% CI)	*P*	HR (95% CI)	*P*	HR (95% CI)	*P*
$PM_{2.5}$	19.83	1.10 (1.04–1.15)	**0.001**	1.09 (1.04–1.15)	**0.001**	1.09 (1.04–1.15)	**0.001**
SO_4^{2-}	4.50	1.12 (1.05–1.20)	**0.001**	1.12 (1.05–1.20)	**0.001**	1.12 (1.05–1.20)	**0.001**
NO_3^-	5.73	1.08 (1.01–1.14)	**0.017**	1.07 (1.01–1.14)	**0.024**	1.07 (1.01–1.14)	**0.023**
NH_4^+	3.92	1.10 (1.03–1.17)	**0.006**	1.09 (1.02–1.17)	**0.009**	1.09 (1.02–1.17)	**0.009**
OM	5.69	1.16 (1.08–1.24)	**< 0.001**	1.15 (1.07–1.23)	**< 0.001**	1.15 (1.07–1.24)	**< 0.001**
BC	1.07	1.21 (1.13–1.31)	**< 0.001**	1.21 (1.12–1.30)	**< 0.001**	1.21 (1.12–1.30)	**< 0.001**

P < 0.05 indicated statistical significance and font was bolded

Abbreviations: *HR* Hazard ratio, *CI* Confidence interval, $PM_{2.5}$ Particulate matter ≤ 2.5 μm in aerodynamic diameter, SO_4^{2-} Sulfates, NO_3^- Nitrates, NH_4^+ Ammonium, *OM* Organic matter, *BC* Black carbon

slightly stronger among participants aged > 60 years, current smokers, non-drinkers, and those with hypertension or dyslipidemia. Elevated risks were also observed in participants with primary- or middle-school education and those in the fourth expenditure quintile (Q4).

For instance, per 1 IQR increase in SO_4^{2-} was associated with a 22% higher risk of VI among adults aged > 60 years (HR = 1.22, 95% CI 1.09–1.36, P < 0.001), and per 1 IQR increase in BC was associated with a 41% higher risk in the Q4 expenditure group (HR = 1.41, 95% CI 1.20–1.65, P < 0.001). However, no statistically significant effect modification was detected for any subgroup (*P* for interaction > 0.05) (Table S9 in Supplementary Material).

Overall, the results based on the 1-year exposure window were in agreement with those from the 3-year analyses, underscoring the robustness of the observed associations between $PM_{2.5}$ components and VI risk.

Discussion

This study, based on a nationally representative longitudinal cohort and incorporating high-resolution air pollution exposure data, systematically evaluated the association between $PM_{2.5}$ and its constituents––including SO_4^{2-}, NO_3^-, NH_4^+, OM, and BC––and the risk of VI. Using time-dependent Cox models, we found that higher ambient $PM_{2.5}$ concentrations were significantly associated with increased risk of VI. Notably, BC and OM exhibited the strongest risk effects, with nonlinear exposure–response relationships. These associations were particularly pronounced among older adults, urban residents, and individuals with lower educational attainment.

These results are consistent with those of previous research. For instance, a large cross-sectional study found that higher levels of $PM_{2.5}$ and its constituents were associated with increased risk of VI in school-aged children, with BC showing the most significant effect. The same study reported nonlinear exposure–response relationships for SO_4^{2-}, NO_3^-, NH_4^+, OM, and BC [25].

VI is a result of multiple ocular diseases, including corneal disorders, cataracts, glaucoma, myopia, and age-related macular degeneration. Numerous epidemiological and experimental studies have suggested links between air pollution and these ocular conditions. For example, Mokhtar et al. reported that long-term $PM_{2.5}$ exposure was associated with decreased corneal nerve fiber density, total nerve length, and fractal dimension, indicating corneal neurodegeneration [26]. Additionally, high $PM_{2.5}$ exposure has been linked to corneal epithelial defects, thinning of the epithelial layer, and limbal stem cell loss [27]. Li et al. demonstrated that each 10 $\mu g/m^3$ increase in $PM_{2.5}$ concentration in the 1, 2, and 3 years before follow-up was associated with HRs of 1.165, 1.138, and 1.091, respectively, for age-related cataracts [13]. A retrospective cohort study in Taiwan involving 97,306 children found that children in the highest quartile of $PM_{2.5}$ exposure (> 41.2 $\mu g/m^3$) had significantly increased risk of developing myopia compared to those in the lowest quartile (< 29.5 $\mu g/m^3$, HR = 1.76) [28]. Additionally, elevated $PM_{2.5}$ levels were associated with an increased risk of glaucoma, characterized by reduced macular ganglion cell-inner plexiform layer thickness, with evidence of a dose–response relationship [16]. Animal studies further support these findings; acute $PM_{2.5}$ exposure induced choroidal vessel dilation and disruption of retinal pigment epithelium (RPE) tight junctions, increasing the risk of retinal edema. Prolonged exposure led to persistent choroidal inflammation and AMD-like degenerative changes in the retina [12].

This study also provides novel insights into the differential toxicity of specific $PM_{2.5}$ constituents. Among them, BC and OM demonstrated the most substantial associations with VI. BC is one of the smallest and most stable carbonaceous fractions of $PM_{2.5}$, primarily emitted from combustion sources such as vehicle exhaust and agricultural burning. Due to its ultrafine nature, BC can penetrate the ocular surface and trigger local inflammatory responses. Experimental studies have shown that BC exposure leads to corneal epithelial cell shedding, inflammatory cell infiltration, and ocular surface microenvironmental imbalance [29]. At the cellular level, BC induces apoptosis in corneal epithelial cells and activates the NLRP3 inflammasome, resulting in increased expression of inflammatory mediators such as IL-1β, NLRP3, caspase-1, and ASC [30]. Systemically, BC exposure may activate mast cells and amplify oxidative stress and systemic inflammation [31]. A study incorporating human genetic data found that the association between BC and elevated intraocular pressure (IOP) was stronger among individuals with high oxidative stress genetic risk scores, suggesting potential gene–environment interactions [32].

OM is one of the most chemically complex $PM_{2.5}$ fractions, comprising polycyclic aromatic hydrocarbons and volatile organic compounds. Due to their lipophilic nature, these compounds can penetrate the eye via the cornea or sclera and elicit toxicological responses. Magnuson et al. demonstrated in a zebrafish model that PAH exposure downregulated genes involved in ocular development and phototransduction, resulting in impaired neural development, increased apoptosis, and reduced optomotor responses [33]. Volatile organic compounds are also known to cause eye irritation and are a common cause of ocular symptoms in high-exposure environments [34].

In contrast to BC and OM, the inorganic secondary aerosol constituents of $PM_{2.5}$, SO_4^{2-}, NO_3^-, and NH_4^+—are formed from precursor gases such as sulfur dioxide, nitrogen oxides, and ammonia through atmospheric chemical reactions. Although these constituents are less chemically reactive, animal studies have shown that exposure to these ions increases malondialdehyde levels and lipid peroxidation in the liver, suggesting elevated systemic oxidative stress [35], which may contribute to ocular inflammation.

Conclusion

This study evaluated the independent associations between long-term exposure to $PM_{2.5}$ and its constituents and the risk of VI in a nationally representative adult cohort. The results identified BC, OM, and SO_4^{2-} as the constituents most strongly associated with visual risk, providing important directions for future mechanistic research.

Strength and limitation

This study examined the associations between long-term exposure to $PM_{2.5}$ and its constituents and the risk of VI in a nationally representative cohort. It identified specific components—particularly BC, OM, and SO_4^{2-}—most strongly linked to VI.

Several limitations should be noted. VI was self-reported rather than clinically measured, which may cause non-differential misclassification. Such random error would generally bias the estimated associations toward the null, leading to an underestimation of the true effects. Residential information was available only at the city level, and exposure differences from living near major industrial areas or busy roads could not be captured. Because VI was assessed every 2–3 years, exact onset times could not be determined, and some temporal misalignment may remain.

Despite these limitations, our findings provide population-based evidence that long-term $PM_{2.5}$ exposure and its components may adversely affect visual health.

Abbreviations

$PM_{2.5}$ (Fine particulate matter: particulate matter with an aerodynamic diameter ≤ 2.5 μm);

SO_4^{2-}	Sulfates;
NO_3^-	Nitrates;
NH_4^+	Ammonium;
OM	Organic matter;
BC	Black carbon;
CHARLS	China Health and Retirement Longitudinal Study;
VI	Visual impairment;
TAP	Tracking Air Pollution;
HR	Hazard ratios;
CI	Confidence interval;
K	Infection point;
RCS	Restricted cubic spline

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-025-25648-y.

Supplementary Material 1.

Acknowledgements

We gratefully acknowledge the teams of the China Health and Retirement Longitudinal Study (CHARLS) and the Tracking Air Pollution in China (TAP) project for providing data. Their contributions made this research possible.

Authors' contributions

Songtao Wang: methodology, software, visualization, validation and original draft preparation; Xueying Li: data curation, visualization, formal analysis and original draft preparation; Xiufen Liu: data curation and original draft preparation; Yibo Wang: investigation and original draft preparation; Dandan Zhou: conceptualization, supervision, validation and review and editing; Chengwei Lu: funding acquisition, project administration, supervision, conceptualization and review and editing. All authors read and approved the final manuscript.

Funding

This research was funded by the National Natural Science Foundation of China under Grant (grant number 82471045, Chengwei Lu).

Data availability

The data used in this study can be found in [https://tapdata.org] (https://tapdata.org) and https://charls.pku.edu.cn.

Declarations

Ethics approval and consent to participate

The data used in this study were obtained from the China Health and Retirement Longitudinal Study (CHARLS), which is publicly available and de-identified. As this research involved secondary analysis of anonymized data, no additional ethical approval was required.

Competing interests

The authors declare no competing interests.

Received: 15 September 2025 / Accepted: 11 November 2025
Published online: 30 December 2025

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